

ROG annotation layers and tags

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1 Introduction

This document describes annotation layers and tags of the training corpus of spoken Slovenian ROG [13].

ROG corpus consists of:

ROG-SST, which includes selected Gos 2.1 transcriptions with manually assigned lemmas and morphosyntactic tags according to the MULTTEXT-East annotation scheme, as well as annotations according to the Universal Dependencies annotation scheme (i.e. part-of-speech categories, morphological features and syntactic dependencies). ROG-SST is distributed as CONLL-U format¹ (2014-2024) (.conllu files).

ROG-Art, which includes all the annotation layers from the ROG-SST plus additional annotation layers for prosodic units, disfluencies and dialogue acts. ROG-Art is distributed as EXMARaLDA format (.EXB files) [7]. In addition to the .EXB files, .EXS files and Rog-Art.coma file are added for searching through the annotated corpus in the EXMARaLDA EXAKT concordancer, .TRS files [2] are added for viewing the transcriptions without annotations in the Transcriber, and .textGrid files [3] with additional prosodic annotations in the Praat (TeG folder) can be found in the ROG-Art folder.

This documentation provides information on annotation layers and tags for both ROG-SST and ROG-Art subset.

2 Data sources and file naming convention

Both ROG-SST and ROG-Art consist of data published in Gos 2.1 corpus [11]², which represents a merge of the:

- Gos 1 [12]³ and
- selected data from the Artur 1 corpus [10]⁴.

¹<https://universaldependencies.org/format.html>

²<http://hdl.handle.net/11356/1863>

³<http://hdl.handle.net/11356/1863>

⁴<http://hdl.handle.net/11356/1772>

Files preserve Gos 2.1 naming, but in order to enhance clarity and provide additional context, prefixes were added to the file names to make them more informative:

FILE NAME PREFIX	DESCRIPTION
Rog-Go1	The files correspond to the original version of the Spoken Slovenian Treebank[5], which was created by selecting data from the GOS1 corpus based on proportional criteria. However, 22 files that were expanded and renamed using the Rog-Go2 naming convention are excluded here.
Rog-Go2	Twenty-two manually selected files containing speech samples, which were excluded from the Rog-Go1 subset, were expanded by adding additional data from the same speech events from the GOS1 corpus.
Rog-Art	The speeches are excerpts taken from the Artur speech corpus.

3 Capitalization, punctuation, segmentation

3.1 Capitalization and punctuation

FILE NAME PREFIX	DESCRIPTION OF CAPI- TALIZATION AND PUNC- TUATION in ROG	History of capitalization and punctuation in the original corpora (GOS 1, GOS 2.1 and/or ARTUR)
Rog-Go1 and Rog-Go2	All three subsets (Rog-Go1, Rog-Go2 and Rog-Art) follow identical manually checked capitalization and punctuation principles: lowercased transcriptions (except for proper nouns), and written-like punctuation, which includes both utterance-final punctuation, such as full stops and question marks, and utterance-medial punctuation, such as commas. While punctuation in Rog-Art remains identical as in the original ARTUR database, it had to be manually added to Rog-Go1 and Rog-Go2 data (see right).	Rog-Go1 and Rog-Go2 transcriptions in the original GOS 1 corpus follow the same truecasing convention (capital letters only for proper names), but a more limited set of punctuation (?, !, and ...). The same principles were preserved, when extending GOS 1 to versions GOS 2.0 and GOS 2.1.
Rog-Art	Transcription and punctuation principles are identical in both CONLL-U and EXB versions of the ROG data.	Rog-Art transcriptions in the original ARTUR database adhered to standard orthographic conventions for capitalization and punctuation. This includes capitalizing both proper names and sentence-initial words, and using both sentence-medial and sentence-final punctuation. When adding ARTUR data to GOS 2.0, punctuation was removed, but then reintroduced in GOS 2.1, together with automatic conversion to truecase capitalization.

3.2 Segmentation

FILE NAME PREFIX	DESCRIPTION OF SEGMENTATION in ROG	History of segmentation in the original corpora (GOS 1, GOS 2.1 and/or ARTUR)
Rog-Go1 and Rog-Go2	Segmentation in Rog-Go1 and Rog-Go2 has been inherited from the original GOS 1 and 2.1 corpus (see right) and remained unchanged. The sentence IDs are identical to those in GOS 1, ensuring long-term traceability and mapping between the datasets.	GOS 1 was manually segmented into utterances, broadly defined as a ‘syntactically, semantically and prosodically completed unit’. The same segmentation was also preserved in GOS 2.0 and 2.1. ^a
Rog-Art	The segmentation of Rog-Art data in EXB format does not feature any explicit segmentation, as all words are on the same level, direct children of <tier> elements. In CONLL-U, Rog-Art has been split into utterances (written-like sentences) to align with the segmentation of Rog-Go1 and Rog-Go2 (see above). This was performed by merging the short segments from the original corpus (see right) based on punctuation marks. Specifically, consecutive segments within each utterance were combined into a single segment until a punctuation mark ‘.’, ‘?’, or ‘...’ was encountered at the end of the first segment, ensuring the segmentation better reflected syntactic structures. These merged sentence units are traceable through sentence ID’s.	The manual segmentation in the original Artur database, which was also propagated to Gos 2.0 and Gos 2.1, was mostly prosody-based, i.e. introducing a segment split after each pause, and did not necessarily align with syntactic unit boundaries, such as syntactically complete phrases or sentences. ^b

^a“Izjava je osnovna enota govora, ki približno ustreza pojmu povedi v pisnem jeziku. Določimo jo tako, da je prozodično, semantično in skladenjsko zaokrožena enota. Pri zelo strnjenem govoru smo pozorni, da izjavo označimo na takem mestu, da je v signalu mogoče nedvoumno označiti mejo med izjavama, ne da bi pri tem odrezali konec prejšnje ali začetek nove izjave. Kjer zgornji vodili omogočata različne odločitve, imajo prednost krajše enote.” [1]

^b“Pomembno je, da segmenti niso predolge enote in da je tam, kjer naredimo mejo segmenta, dovolj premora v govoru, da lahko določimo mejo segmenta, ne da odrežemo del predhodnega ali del naslednjega fonema. /.../ Okvirno velja pravilo, da naj bo premor dolg vsaj 0,2 sekunde, lahko pa tudi manj v naslednjih primerih: (1) če bi bili sicer segmenti predolgi, tj. več kot okoli 10 sekund, (2) če se menjajo govorci ali (3) če bi v segment s hkratnim govorom tako zajeli tudi veliko nekratnega govora. Vedno pa mora biti med segmenti vsaj malo premora, da je meja med besedami vidna tudi na signalu. /.../ Idealno je, če lahko segment sovпада s stavki ali kratkimi povedmi oz. s semantično in skladenjsko kolikor mogoče zaključenimi, smiselnimi enotami, vendar pogosto to ni mogoče. Glavni vodili za meje med segmenti sta zato: • kratek premor v govoru in • dolžina segmenta, ki naj ne bo predolga, tj. več kot okoli 10 sekund.” [9]

4 Lemmas

Lemmas were first assigned automatically using the CLASSLA-Stanza model for lemmatisation of standard Slovenian 2.0 [8] and then manually validated/corrected. The manual corrections followed the lemmatization guidelines available at: <https://wiki.cjvt.si/books/05-lemmatization>.

5 Universal part-of-speech categories (upos)

Universal part-of-speech categories (UPOS column in ROG-SST files) follow the Universal Dependencies annotation scheme, documented in the following guidelines:

- general UD guidelines for part-of-speech (in English, online): <https://universaldependencies.org/u/pos/index.html>
- general UD guidelines for Slovenian (in English, online): <https://universaldependencies.org/sl/>
- detailed UD guidelines for Slovenian (in Slovene, standalone document), v1.7: <https://wiki.cjvt.si/books/07-universal-dependencies/page/oznacevalne-smernice>

Part-of-speech tags used:

TAG	DESCRIPTION
ADJ	adjective
ADP	adposition
ADV	adverb
AUX	auxiliary
CCONJ	coordinating conjunction
DET	determiner
INTJ	interjection
NOUN	noun
NUM	numeral
PART	particle
PRON	pronoun
PROPN	proper noun
PUNCT	punctuation
SCONJ	subordinating conjunction
SYM	symbol
VERB	verb
X	other

As described by Dobrovoljc [4], a semi-automatic procedure was employed to produce UD morphological annotation, by which the manually validated Multext-East (JOS) morphosyntactic tags (see below) have been converted to UD parts-of-speech and feature combinations with the `jos2ud`⁵ conversion script. Mistakes and inconsistencies identified during subsequent dependency annotation and data validation have been manually corrected.

⁵<https://github.com/clarinsi/jos2ud>

6 MULTEXT-East morphosyntactic tags (xpos)

Morphosyntactic tags in ROG-Art and ROG-SST were assigned automatically using the The CLASSLA-Stanza model for morphosyntactic annotation of standard Slovenian 2.0⁶ and then manually validated/corrected. The tags follow the Slovene Multext-East Specifications (v6)⁷.

7 Universal morphological features (feats)

Universal morphological features (FEATS column in ROG-SST files) follow the Universal Dependencies annotation scheme, documented in the following guidelines:

- general UD guidelines for morphological features (in English, online): <https://universaldependencies.org/u/feat/index.html>
- general UD guidelines for Slovenian (in English, online): <https://universaldependencies.org/sl/>
- detailed UD guidelines for Slovenian (in Slovene, standalone document), v1.7: <https://wiki.cjvt.si/books/07-universal-dependencies/page/oznacevalne-smernice>

Tags for morphological features used (found as feature-value pairs, e.g. Tense=Pres):

Feature	Values	Description
Abbr	Yes	abbreviation
Animacy	Anim, Inanim	animacy
Aspect	Imp, Perf	aspect
Case	Nom, Gen, Dat, Acc, Loc, Ins	case
Definite	Ind, Def	definiteness or state
Degree	Pos, Cmp, Sup	degree
Foreign	Yes	foreign word
Gender	Masc, Fem, Neut	gender
Gender[psor]	Masc, Fem, Neut	possessor's gender
Mood	Ind, Imp, Cnd	mood
Number	Sing, Dual, Plur	number
Number[psor]	Sing, Dual, Plur	possessor's number
NumForm	Word, Digit, Roman	numeral form
NumType	Card, Ord, Mult, Sets	numeral type
Person	1, 2, 3	person
Polarity	Neg, Pos	polarity
Poss	Yes	possessive
PronType	Prs, Int, Rel, Dem, Tot, Neg, Ind	pronominal type
Reflex	Yes	reflexive
Tense	Pres, Fut	tense
Variant	Bound, Short	alternative form of word
VerbForm	Fin, Inf, Sup, Part, Conv	form of verb or deverbative

As described by Dobrovoljc [4], a semi-automatic procedure was employed to produce UD

⁶<http://hdl.handle.net/11356/1767>

⁷<https://nl.ijs.si/ME/V6/msd/html/msd-sl.html>

morphological annotation, by which the manually validated Multext-East (JOS) morphosyntactic tags (see below) have been converted to UD parts-of-speech and feature combinations with the `jos2ud`⁸ conversion script. Mistakes and inconsistencies identified during subsequent dependency annotation and data validation have been manually corrected.

8 Universal syntactic relations (head and deprel)

The annotation of syntactic relations between words (HEAD and DEPREL columns in ROG-SST files) follows the Universal Dependencies annotation scheme, documented in the following guidelines:

- general UD guidelines for syntactic relations (in English, online): <https://universaldependencies.org/u/dep/index.html>
- general UD guidelines for Slovenian (in English, online): <https://universaldependencies.org/sl/>
- detailed UD guidelines for Slovenian (in Slovene, standalone document), v1.7: <https://wiki.cjvt.si/books/07-universal-dependencies/page/oznacevalne-smernice>

Tags for dependency relations used (without subtypes of the type `cc:preconj`):

Tag	Description
<code>acl</code>	clausal modifier of noun
<code>advcl</code>	adverbial clause modifier
<code>advmod</code>	adverbial modifier
<code>amod</code>	adjectival modifier
<code>appos</code>	appositional modifier
<code>aux</code>	auxiliary verb
<code>case</code>	case marking preposition
<code>cc</code>	coordinating conjunction
<code>ccomp</code>	clausal complement
<code>conj</code>	conjunct
<code>cop</code>	copula verb
<code>csubj</code>	clausal subject
<code>dep</code>	unspecified dependency
<code>det</code>	determiner
<code>discourse</code>	discourse element
<code>dislocated</code>	dislocated element
<code>expl</code>	expletive
<code>fixed</code>	fixed multi-word expression
<code>flat</code>	flat multi word-expression
<code>goeswith</code>	disjointed token
<code>iobj</code>	indirect object
<code>list</code>	list
<code>mark</code>	marker (subordinating conjunction)

⁸<https://github.com/clarinsi/jos2ud>

Tag	Description
nmod	nominal modifier
nsubj	nominal subject
nummod	numeric modifier
obj	(direct) object
obl	oblique nominal (adjunct)
orphan	dependent of missing parent
parataxis	parataxis
punct	punctuation symbol
reparandum	overridden disfluency
root	root element
vocative	vocative
xcomp	open clausal complement

As described by Dobrovoljc [4], a semi-automatic procedure was employed to produce UD dependency annotation, involving automatically pre-parsed transcriptions with the customized Slovenian Trankit model [6], which were then manually checked by 2-3 independent annotators.

9 Additional

In this tier corrections of transcription are annotated, e.g. when a word or filler in the transcription is missing, e.g. ‘nnn’ means that filler ‘nnn’ is missing in the defined segment.

Furthermore, lengthened phonemes in case of the lengthening are marked with sign (:) after the phoneme, e.g. ‘o(:)d’ means that phoneme ‘o’ in the word ‘od’ is lengthened.

10 Disfluency annotations

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